

Project Title:

Automated Vulnerability Assessment & AI-Assisted Security Reporting Tool

Designed by:

Chandni Jamali

Course Name:

Cybersecurity

Instructors:

Ma’am Pareesa & Ma’am Shiza

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CyberScope

Automated Vulnerability Assessment & AI-Assisted Security Reporting Tool

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1. Introduction

CyberScope is an automated vulnerability assessment and AI-assisted security reporting tool developed using Bash and Python in the Kali Linux environment. The tool is designed to perform reconnaissance, network scanning, and selected web security assessments against a specified target.

It collects assessment results and uses the Gemini API to assist in generating a structured HTML security report. The purpose of CyberScope is to simplify the vulnerability assessment process, organize technical findings, and make the results easier to understand.

2. Problem Statement

Manual vulnerability assessment involves using multiple security tools, collecting their outputs separately, and analyzing the results to prepare a report. This process can be time-consuming and may require considerable manual effort.

CyberScope addresses this problem by integrating selected security assessment tools into an automated workflow and providing an AI-assisted reporting mechanism. It aims to reduce repetitive tasks and organize assessment evidence in one place.

3. Objectives

The main objectives of CyberScope are:

- To automate selected reconnaissance and vulnerability assessment tasks.
- To identify accessible ports and running services.
- To perform selected network and web security checks.
- To collect and organize assessment results automatically.
- To identify potential vulnerabilities and related CVE information where available.
- To use the Gemini API to assist in generating an HTML security report.
- To provide recommendations and a structured summary of assessment findings.

4. System Design and Approach

CyberScope follows a structured workflow in which the user provides a target, and the tool performs the configured assessment tasks.

4.1 Target Input and Execution

The process begins when the user executes CyberScope and provides the target IP address, domain, or URL according to the supported input format. The tool prepares the target for the assessment and starts the configured scanning process.

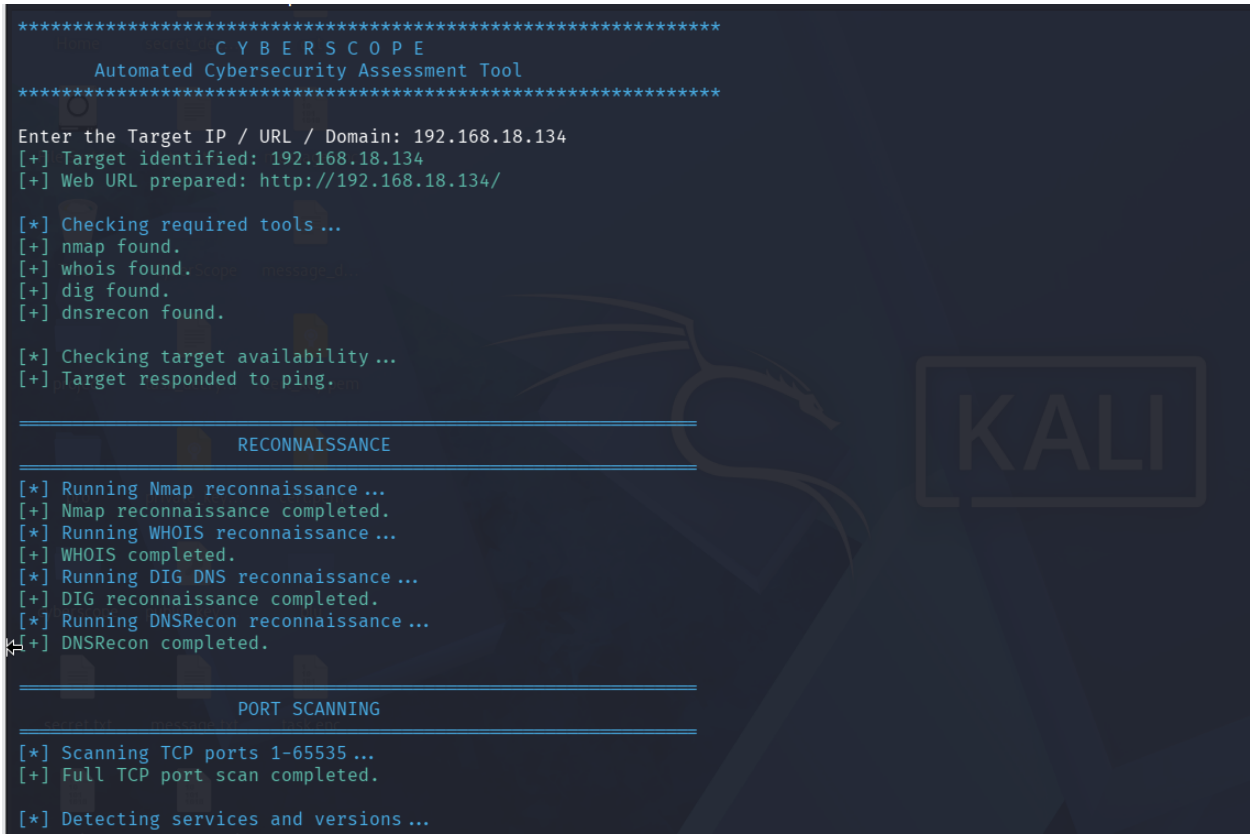


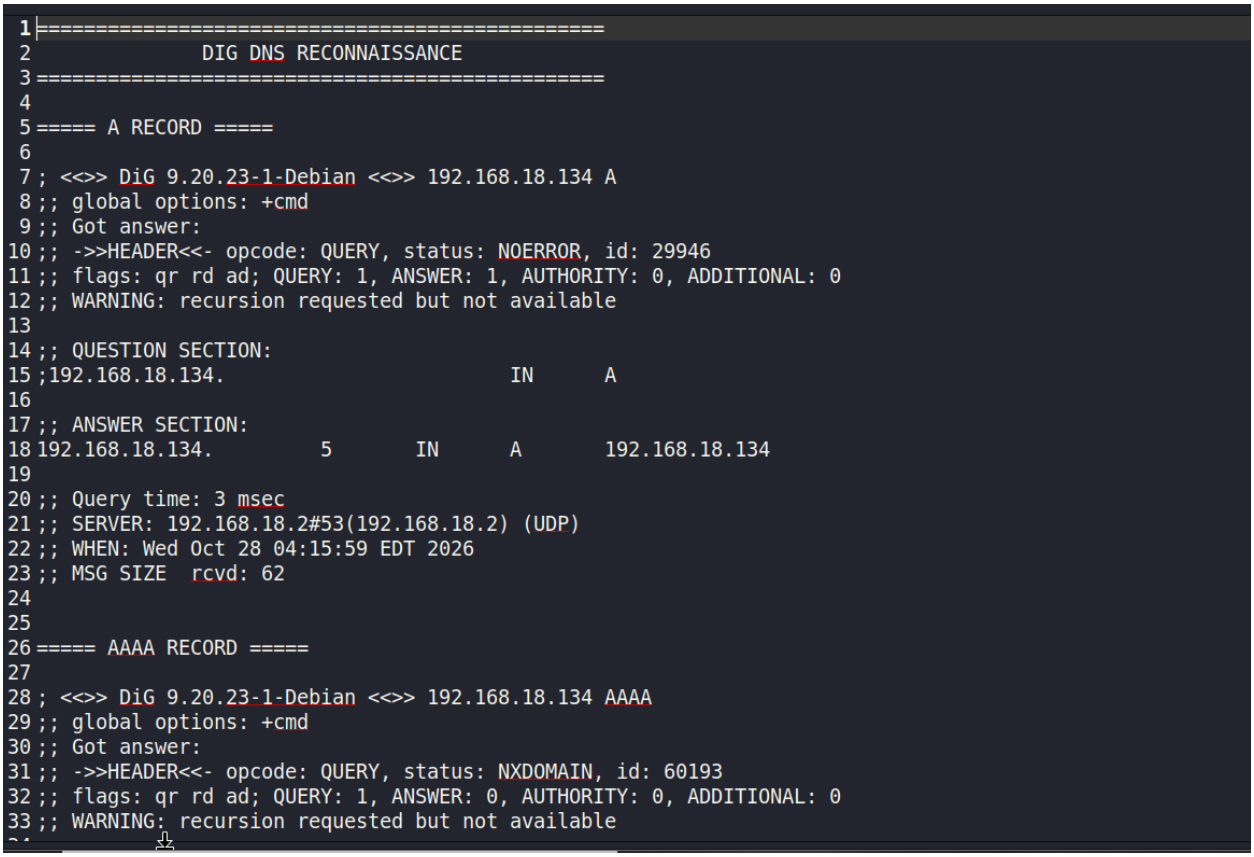
Figure 1: Execution of CyberScope in the Kali Linux terminal. Insert the screenshot showing the tool starting, its banner, and the target input or execution process.

4.2 Reconnaissance

The reconnaissance phase gathers preliminary information about the target. CyberScope uses selected tools and commands, including:

- **WHOIS:** Collects available domain registration information.
- **DIG:** Retrieves DNS records and related information.
- **DNSRecon:** Performs DNS enumeration to identify available DNS information.

The collected information helps establish an initial understanding of the target before further assessment.



```
1 ==============================================
2               DIG DNS RECONNAISSANCE
3 ==============================================
4
5 ===== A RECORD =====
6
7 ; <<>> DiG 9.20.23-1-Debian <<>> 192.168.18.134 A
8 ;; global options: +cmd
9 ;; Got answer:
10 ;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 29946
11 ;; flags: qr rd ad; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 0
12 ;; WARNING: recursion requested but not available
13
14 ;; QUESTION SECTION:
15 ;192.168.18.134.                        IN      A
16
17 ;; ANSWER SECTION:
18 192.168.18.134.         5       IN      A       192.168.18.134
19
20 ;; Query time: 3 msec
21 ;; SERVER: 192.168.18.2#53(192.168.18.2) (UDP)
22 ;; WHEN: Wed Oct 28 04:15:59 EDT 2026
23 ;; MSG SIZE  rcvd: 62
24
25
26 ===== AAAA RECORD =====
27
28 ; <<>> DiG 9.20.23-1-Debian <<>> 192.168.18.134 AAAA
29 ;; global options: +cmd
30 ;; Got answer:
31 ;; ->>HEADER<<- opcode: QUERY, status: NXDOMAIN, id: 60193
32 ;; flags: qr rd ad; QUERY: 1, ANSWER: 0, AUTHORITY: 0, ADDITIONAL: 0
33 ;; WARNING: recursion requested but not available
```

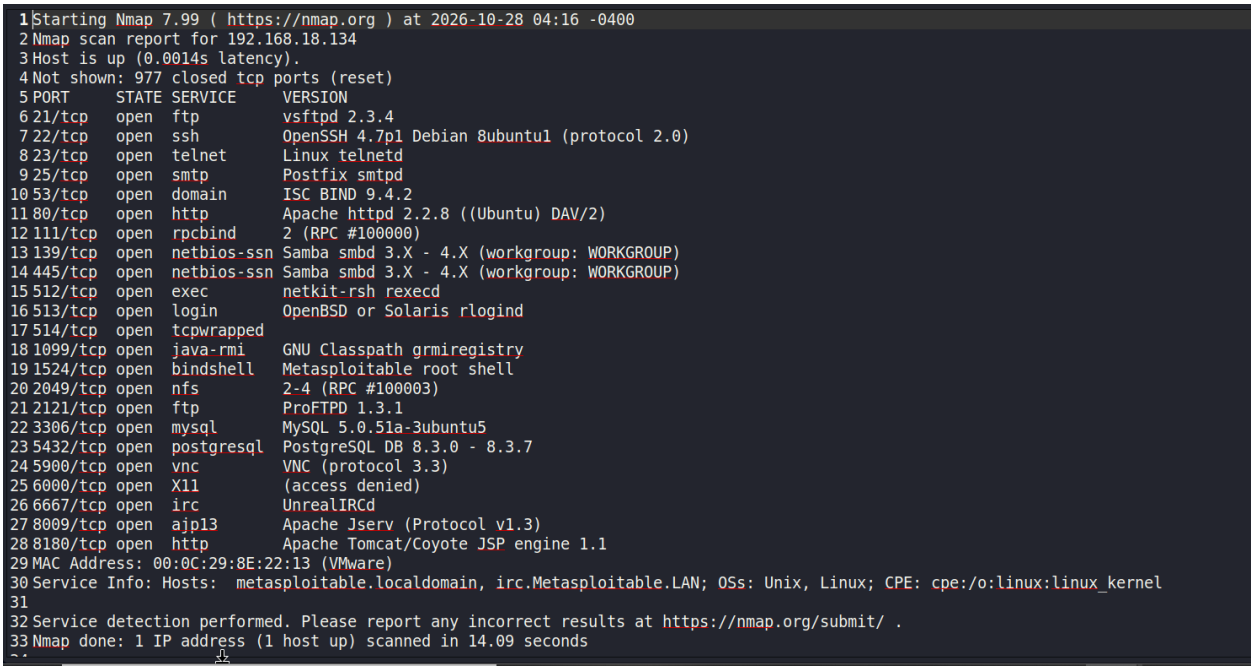
Figure 2: Reconnaissance results generated by CyberScope. Insert the screenshot showing the WHOIS, DIG, DNSRecon, or other configured reconnaissance outputs.

4.3 Network Scanning and Vulnerability Assessment

CyberScope uses Nmap to perform network scanning and service detection. This phase is intended to identify:

- Open ports.
- Running services.
- Service version information.
- Potential vulnerabilities detected through selected Nmap scripts.
- Related CVE information where available.

The results are saved for further analysis and inclusion in the final report.



```
Starting Nmap 7.99 ( https://nmap.org ) at 2026-10-28 04:16 -0400
Nmap scan report for 192.168.18.134
Host is up (0.0014s latency).
Not shown: 977 closed tcp ports (reset)
PORT     STATE SERVICE     VERSION
21/tcp   open  ftp         vsftpd 2.3.4
22/tcp   open  ssh         OpenSSH 4.7p1 Debian 8ubuntu1 (protocol 2.0)
23/tcp   open  telnet      Linux telnetd
25/tcp   open  smtp        Postfix smtpd
53/tcp   open  domain      ISC BIND 9.4.2
80/tcp   open  http        Apache httpd 2.2.8 ((Ubuntu) DAV/2)
111/tcp  open  rpcbind     2 (RPC #100000)
139/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
445/tcp  open  netbios-ssn Samba smbd 3.X - 4.X (workgroup: WORKGROUP)
512/tcp  open  exec        netkit-rsh rexecd
513/tcp  open  login       OpenBSD or Solaris rlogind
514/tcp  open  tcpwrapped
1099/tcp open  java-rmi    GNU Classpath grmiregistry
1524/tcp open  bindshell   Metasploitable root shell
2049/tcp open  nfs         2-4 (RPC #100003)
2121/tcp open  ftp         ProFTPD 1.3.1
3306/tcp open  mysql       MySQL 5.0.51a-3ubuntu5
5432/tcp open  postgresql  PostgreSQL DB 8.3.0 - 8.3.7
5900/tcp open  vnc         VNC (protocol 3.3)
6000/tcp open  X11         (access denied)
6667/tcp open  irc         UnrealIRCd
8009/tcp open  ajp13       Apache Jserv (Protocol v1.3)
8180/tcp open  http        Apache Tomcat/Coyote JSP engine 1.1
MAC Address: 00:0C:29:8E:22:13 (VMware)
Service Info: Hosts:  metasploitable.localdomain, irc.Metasploitable.LAN; OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 14.09 seconds
```

Figure 3: Nmap scanning results. Insert the screenshot showing the port scan, service detection, and vulnerability assessment output.

4.4 Web Security Assessment

CyberScope also includes selected web security assessment modules to examine web-related security issues.

Depending on the configured modules, the assessment may include:

- **DIRB:** Directory and file discovery on a web server.
- **PwnXSS:** Checks for potential cross-site scripting vulnerabilities.
- **SQL Injection Assessment:** A selected practical assessment of SQL injection vulnerabilities in an authorized testing environment.

The results of these checks are collected and stored for reporting.

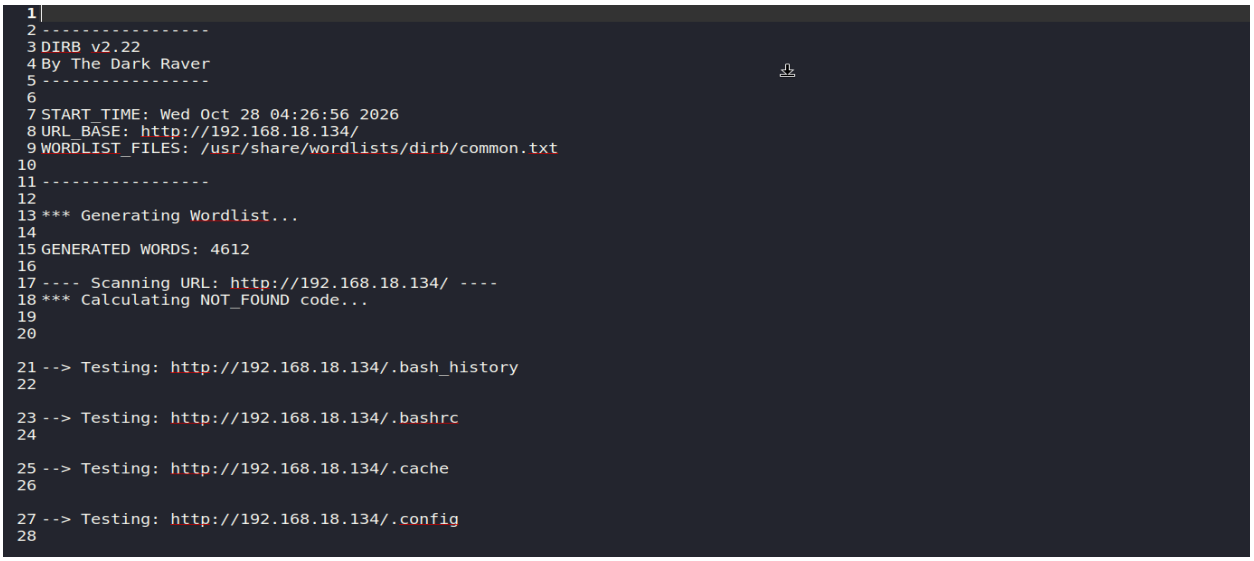


Figure 4: Web security assessment results. Insert the screenshot showing the execution or output of DIRB, PwnXSS, or the configured web security testing module.

4.5 Evidence Collection and AI-Assisted Report Generation

After the assessment, CyberScope organizes the available results in its Results directory. The Python-based reporting component collects the existing evidence, filters unnecessary information, and prepares it for submission to the Gemini API.Gemini is used to assist in generating a structured HTML report containing assessment details, identified findings, risk-related information, and security recommendations based on the available evidence.If AI report generation is unsuccessful, the reporting component can provide a fallback HTML report based on the collected evidence.



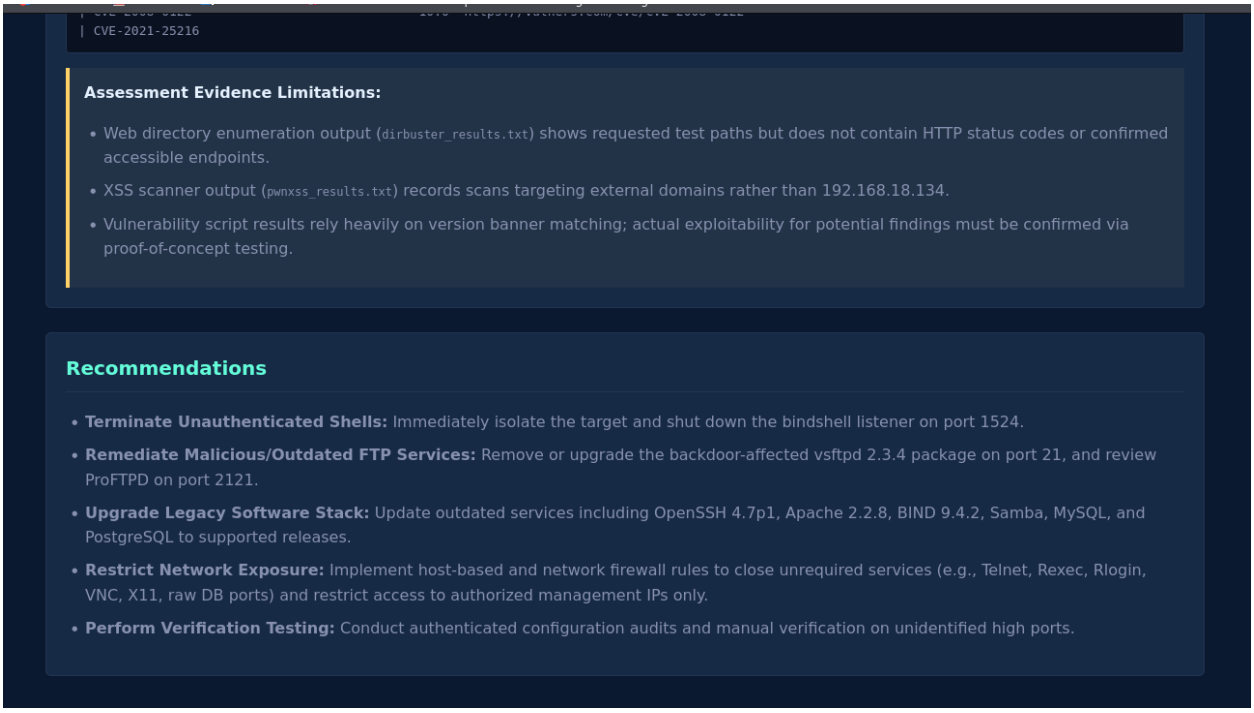


Figure 5: AI-assisted HTML vulnerability assessment report generated by CyberScope. Insert a screenshot of the report opened in a web browser, showing its title and report content.

5. Tools and Technologies Used

The project uses the following tools and technologies:

Tool / Technology	Purpose
Kali Linux	Operating system and testing environment
Bash	Main automation script
Python	Evidence processing and report generation
Nmap	Network scanning and service detection
WHOIS	Domain information gathering
DIG	DNS information gathering
DNSRecon	DNS enumeration
DIRB	Web directory discovery
PwnXSS	Web security assessment
Gemini API	AI-assisted report generation
HTML/CSS	Final report presentation

Note: The tools and modules listed above represent the project's configured components. Include only those successfully implemented and tested in the final submitted version.

6. Project Workflow

The general workflow of CyberScope is:

1. User provides the target.
2. The tool performs target preparation.
3. Reconnaissance gathers preliminary information.
4. Nmap performs network and vulnerability assessment.
5. Configured web security modules perform selected checks.
6. Assessment evidence is stored in the Results directory.
7. The Python reporting component processes the collected evidence.
8. Gemini assists in generating the HTML report.
9. The final report is saved for review.

7. Implementation and Testing

7.1 Development Environment

CyberScope was developed and tested in Kali Linux. Bash was used to automate the assessment workflow, while Python was used for evidence processing and HTML report generation.

7.2 Assessment Execution

The tool was executed against a designated testing target. The configured reconnaissance, network scanning, and web security assessment components were used to collect information.

7.3 AI Report Generation

The reporting component was tested using the collected assessment evidence. The Gemini API successfully generated an HTML report.

During the successful test, the target IP address was passed to the reporting script, and the API returned a completion status of STOP. The generated report was saved at:

Results/AI_Vulnerability_Assessment_Report.html

7.4 Project Files and Output Organization

The project uses a structured directory to organize its scripts and assessment results. The terminal directory listing demonstrates the organization of the project files and generated outputs.



Figure 6: CyberScope project directory structure. Insert a screenshot of the Kali Linux terminal showing the `ls` command and the project files and folders, such as `script.sh`, `ai_report.py`, `README.md`, and `Results`.

8. Results and Observations

The implementation demonstrates the integration of selected security assessment tools into a common workflow. The tool collects assessment evidence and stores it for further processing.

The AI-assisted reporting component successfully generated an HTML report using the Gemini API and the available assessment evidence.

The actual ports, services, vulnerabilities, CVEs, and risk levels identified during testing should be documented using the corresponding scan outputs and generated report.

9. Challenges and Solutions

During development, several technical challenges were encountered:

- **Dependency Management:** Required tools and Python libraries needed to be available in the Kali Linux environment.
- **Target Handling:** Target information needed to be passed correctly between the Bash and Python components.
- **Evidence Processing:** Large scan outputs required filtering and organization before being submitted to the API.
- **API Response Limitations:** Response generation needed appropriate configuration to avoid incomplete output.
- **HTML Report Generation:** The generated response needed validation to ensure that it was a usable HTML document.

These challenges were addressed through script corrections, improved evidence handling, API configuration, and report validation.

10. Limitations

CyberScope has some limitations:

- Assessment results depend on the availability and configuration of the selected tools.
- Some security checks may take additional time depending on the target.

- Results depend on the accessibility and responsiveness of the target system.
- AI-generated descriptions and recommendations require human review.
- The tool does not guarantee detection of every vulnerability.

11. Future Improvements

Possible future improvements include:

- Adding more vulnerability detection modules.
- Improving scan performance and evidence filtering.
- Enhancing HTML report design and visualization.
- Adding support for additional target formats.
- Improving vulnerability verification and risk assessment.
- Extending the reporting system with more detailed remediation guidance.

12. Lessons Learned

The development of CyberScope provided practical experience in Bash scripting, Python programming, Linux environments, network reconnaissance, vulnerability assessment, and API integration.

It also demonstrated the importance of automation, proper evidence collection, error handling, and clear technical reporting in cybersecurity projects.

13. Conclusion

CyberScope demonstrates how selected cybersecurity tools can be integrated into an automated vulnerability assessment workflow. By combining Bash automation, Python-based evidence processing, and Gemini-assisted HTML report generation, the project provides a structured approach to collecting and presenting security assessment results.

The project also provides practical learning experience in cybersecurity automation and demonstrates the potential of AI-assisted reporting as a supporting component of security assessment.
